



1 of 1)

Queen Adedire (Attempt

Test Information

QUESTION 1: CALCULATED NUMERIC

25

out of 25 points

Refer to page 334 of Chapter 7 that is uploaded on the UIC Blackboard in the Course Textbook content area.

1. Identify the error(s) in the calculation of the Dew Point pressure
2. Propose a detailed algorithm for the correct calculation of the Dew Point pressure.



3. Carry out the calculation and enter your result in **bar** in the textbox below. Round your numerical answer into **two decimal digits**.

Given Answer: 3.70

Correct Answer: 3.7

Answer range +/- 0 (3.7 - 3.7)

QUESTION 2: FILE RESPONSE

75

out of 75 points

Refer to page 334 of Chapter 7 that is uploaded on the UIC Blackboard in the Course Textbook content area.

1. Identify the error(s) in the calculation of the Dew Point pressure
2. Propose a detailed algorithm for the correct calculation of the Dew Point pressure.
3. Carry out the calculation and enter your result in **bar** in the textbox below. Round your numerical answer into **two decimal digits**.
4. **Prepare a detailed technical report of your work and upload it by the due date and time: Monday December 1st, 11:59PM**

Given

At the Dew point the mixture is almost a hundred percent vap.pdf

$$\sum z_j K_j = 1 \Rightarrow \sum z_j \gamma_i \frac{P_j^{sat}}{P_{BUBBLE}} = 1 \Rightarrow \boxed{P_{BUBBLE} = \sum z_j \gamma_i P_j^{sat}}$$

Step 2: We now calculate the liquid activity coefficients γ_1, γ_2 at the temperature and composition of the mixture. The values we get from the CHE Calculator are: $\gamma_1 = 1.855753$ and $\gamma_2 = 1.107145$. Next, we calculate the saturated vapor pressures $P_1^{sat} = 8.2626 \text{ bar}$ and $P_2^{sat} = 1.7051 \text{ bar}$ at the bubble point temperature and carrying out the calculation for the bubble point pressure we obtain

$$P_{BUBBLE} = \sum z_j \gamma_i P_j^{sat} \Rightarrow P_{BUBBLE} = 6.6965 \text{ bar}$$

Step 3: First, we need to calculate the dew point pressure of the liquid mixture using the Rachford-Rice Equation with $\alpha = 1$. The equation gives,

$$\sum \frac{z_j}{K_j} = 1 \Rightarrow \sum \frac{z_j}{\gamma_i \frac{P_j^{sat}}{P_{DEW}}} = 1 \Rightarrow \boxed{P_{DEW} = \left[\sum \frac{z_j}{\gamma_i P_j^{sat}} \right]^{-1}}$$

Step 4: We now calculate the liquid activity coefficients γ_1, γ_2 at the temperature and composition of the mixture. The values we get from the CHE Calculator are the same ones calculate in the Step 2 above, i.e., $\gamma_1 = 1.855753$ and $\gamma_2 = 1.107145$. Similarly, we use the saturated vapor pressures $P_1^{sat} = 8.2626 \text{ bar}$ $P_2^{sat} = 1.7051 \text{ bar}$ at the dew point temperature and carrying out the calculation for the dew point pressure we obtain

$$P_{DEW} = \left[\sum \frac{z_j}{\gamma_i P_j^{sat}} \right]^{-1} \Rightarrow P_{DEW} = 3.948 \text{ bar}$$

At the Dew point the mixture is almost a hundred percent vapor and about to form a drop of liquid. At this point $z_i = y_i$ however $z_i \neq x_i$ because when the liquid first forms it does not have the same composition of vapor, the liquid activity coefficient as calculated on page 334 is incorrect, therefore the dew point pressure is incorrect.

Assumptions

- Ideal gas
- Non-ideal Liquid $K_i = \frac{y_i}{x_i} = \gamma_i \frac{P_i^{sat}}{P}$

To find the correct value for the dew point pressure the procedure is as follows

Step 1: In the data base add ethanol and toluene as components and add the BIP as follows

Aij: 2.5417

Aji: -1.1016

Bij: -1310.7500

Bji: 28.6656